AIME Combinatorics

Albert Cao (abeot)

January 25, 2026

This handout assumes that you are comfortable qualifying for AIME, or if not, comfortable solving the early end of AIME problems (first 5, for example).

Should be pretty comprehensive; mainly created because I told someone that "AIME combo is literally just the same ideas over and over again".

§1 The Basics

It is required to know what combinations and permutations are, as well as other counting techniques. Casework, complementary counting, and constructive counting are just a few to name. These techniques are for more basic problems.

Example 1.1 (2022 AIME I Problem 9)

Ellina has twelve blocks, two each of red (**R**), blue (**B**), yellow (**Y**), green (**G**), orange (**O**), and purple (**P**). Call an arrangement of blocks even if there is an even number of blocks between each pair of blocks of the same color. For example, the arrangement

$$\mathbf{R\ B\ B\ Y\ G\ G\ Y\ R\ O\ P\ P\ O}$$

is even. Ellina arranges her blocks in a row in random order. The probability that her arrangement is even is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

This might look like quite a daunting problem, as it is number 9. The idea (which one might be able to find) is to consider spots with an even position and spots with an odd position (if you number the spots). However, consider how we can place the first R, then the second R, then the first B, etc. and divide by 2^6 at the end for overcounting, and the number of total possible ways is also easy to find! (Note: was it necessary to divide by 2^6?)

Example 1.2 (2018 AIME II Problem 11)

Find the number of permutations of $1, 2, 3, 4, 5, 6$ such that for each k with $1 \le k \le 5$, at least one of the first k terms of the permutation is greater than k.

Solution. Because it is annoying to deal with cases where the first few numbers satisfy all the conditions, we complementary count, where for at least 1 value of k, the first k terms are less than or equal to k.

We perform casework on each of the k, which is left as an exercise, and summing and subtracting from 720, we obtain $\boxed{461}$. □

Exercise 1.3 (2023 AIME I Problem 1)**.** Five men and nine women stand equally spaced around a circle in random order. The probability that every man stands diametrically opposite a woman is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Also, for some problems it is good to know PIE (Principle of Inclusion Exclusion), especially if complementary counting or something like that. The next exercise shows how it can be used.

Theorem 1.4 (PIE)

If $(A_i)_{1\leq i\leq n}$ are finite sets, then:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n-1} |A_1 \cap \cdots \cap A_n|$$

Exercise 1.5 (2020 AIME II Problem 9)**.** While watching a show, Ayako, Billy, Carlos, Dahlia, Ehuang, and Frank sat in that order in a row of six chairs. During the break, they went to the kitchen for a snack. When they came back, they sat on those six chairs in such a way that if two of them sat next to each other before the break, then they did not sit next to each other after the break. Find the number of possible seating orders they could have chosen after the break.

§2 Counting In Two Ways (aka Bijections)

§2.1 Combinatorial Identities

I'll just dump everything that needs to be known here.

Lemma 2.1 (Stars and Bars)

To but n balls (indistinguishable) into k bins (distinguishable) is $\binom{n+k-1}{k-1}$.

Proof. Create a bijection between stars and bars a line, with bars representing barriers between bins. □

Lemma 2.2

$$\binom{n}{k} = \binom{n}{n-k}$$

Theorem 2.3 (Hockey Stick Identity)

$$\binom{k}{k} + \binom{k+1}{k} + \cdots + \binom{n}{k} = \binom{n+1}{k+1}$$

Proof. Suppose we need to choose $k+1$ numbers out of the first $n+1$ natural numbers. Casework on the least number gives the LHS. □

Theorem 2.4 (Vandermonde's Identity)

$$\sum_{i=0}^{k} \binom{n}{i}\binom{m}{k-i} = \binom{n+m}{k}$$

Proof. Left as an exercise. □

§2.2 Individual vs Group

This is about the idea of observing how each item can contribute to a sum. Also gives a few more combinatorial identities. The proofs here also use an idea of "committee selection", where $\binom{n}{k}$ is the same as choosing a k member committee from n people.

Example 2.5

How many subsets can be formed from a set with n elements?

Solution. Counting by each element and whether or not it can be in the subset gives 2^n. Note that counting by the size of subsets also gives that

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$$

□

Lemma 2.6 (Committee Leader)

For a positive integer n, show that

$$\sum_{i=0}^{n} i \cdot \binom{n}{i} = n2^{n-1}$$

Proof. The left-hand side does casework by the size of committees and then chooses a leader; the RHS looks at the group as a whole, first choosing the leader and then looking at each person. □

Exercise 2.7. Show that the number of ways to choose two subsets B and A such that B is a subset of A and A is a subset of S is 3^n for a set S with n elements.

Example 2.8 (2022 AIME I Problem 12)

For any finite set X, let $|X|$ denote the number of elements in X. Define

$$S_n = \sum |A \cap B|,$$

where the sum is taken over all ordered pairs (A, B) such that A and B are subsets of $\{1, 2, 3, \ldots, n\}$ with $|A| = |B|$. For example, $S_2 = 4$ because the sum is taken over the pairs of subsets

$$(A, B) \in \{(\emptyset, \emptyset), (\{1\}, \{1\}), (\{1\}, \{2\}), (\{2\}, \{1\}), (\{2\}, \{2\}), (\{1, 2\}, \{1, 2\})\},$$

giving $S_2 = 0 + 1 + 0 + 0 + 1 + 2 = 4$. Let $\frac{S_{2022}}{S_{2021}} = \frac{p}{q}$, where p and q are relatively prime positive integers. Find the remainder when $p + q$ is divided by 1000.

Solution. Casework by the size of subsets doesn't actually get anywhere; however, look at the perspective of each element in the intersection, which automatically takes care of cardinality (subsets with k elements are counted for each of those k elements). This is a prime example of summation swapping. Then the rest is just a binomial sum that can be evaluated with Vandermonde's, so we should get something of the form $n\binom{2n-2}{n-1}$. Complete the solution as an exercise. □

§2.3 Other bijections

Lemma 2.9 (Shortest Paths on a Grid)

The number of shortest paths to move from one corner to the other corner of an m by n grid is $\binom{m+n}{n}$.

Proof. We need to move one direction m times and the other direction n times. Writing this out as a string, where L symbolizes the direction that is m steps long, then there are $\binom{m+n}{n}$ such strings. □

Lemma 2.10 (Separation)

Suppose that we have n available slots. We want to put k elements in the slots, $k \leq \frac{n}{2}$ such that no two of the elements are adjacent. Then the number of ways to arrange the k elements is $\binom{n-k+1}{k}$.

Proof. We first separate the elements by putting a slot in between each, but we still have $n - k - (k - 1) = n - 2k + 1$ slots to distribute. It's naive to assume that the distribution is then just $\binom{n}{2k+1}$, because it's important to notice that it doesn't matter whether or not a slot goes to the left or right of another slot; what actually matters is the length of consecutive slots between each element. So, in other words, we want to put each of the $n - 2k + 1$ slots in $k + 1$ groups of slots, and by stars-and-bars this is just $\binom{n-2k+1+k}{n-2k+1} = \binom{n-k+1}{k}$. □

Don't memorize the statements, but internalize the ideas of the proofs (so the statements are reachable).

§2.4 Pairing (aka Symmetry)

The idea is that one thing corresponds to some complement of it.

Example 2.11

What is the probability of flipping more heads than tails if we flip n coins, with n odd?

Solution. It's intuitively $\frac{1}{2}$, since each case of more heads coressponds to a case with more tails by flipping each coin over. □

Example 2.12

What is the probability of flipping more heads than tails if we flip n coins, with n even?

Solution. Now we need to consider when we flip the same amount of heads and tails; after subtracting, then the rest is still symmetric. □

Exercise 2.13 (HMMT Feb Guts 2011/22)**.** Find the number of ordered triples (a, b, c) of pairwise distinct integers such that $-31 \leq a, b, c \leq 31$ and $a + b + c > 0$.

Exercise 2.14 (1983 AIME Problem 13)**.** For $1, 2, 3, ..., n$ and each of its non-empty subsets a unique alternating sum is defined as follows. Arrange the numbers in the subset in decreasing order and then, beginning with the largest, alternately add and subtract successive numbers. For example, the alternating sum for 1, 2, 3, 6, 9 is 9 - 6 + 3 - 2 + 1 = 5 and for 5 it is simply 5. Find the sum of all such alternating sums for $n = 7$.

Make sure to do small cases to get an idea of what the answer might be, then prove it with pairing.

§2.5 Choices

The idea is to consider "how many choices can I make before the rest is fixed".

Example 2.15

How many ways can n coins be flipped such that an even number of heads come up?

Solution. Notice that the first $n - 1$ flips can be anything, and then the last flip is forced. Thus, the answer is 2^{n-1}. □

A good example is also this easy problem:

Example 2.16 (2018 AMC 10A Problem 20)

A scanning code consists of a 7×7 grid of squares, with some of its squares colored black and the rest colored white. There must be at least one square of each color in this grid of 49 squares. A scanning code is called symmetric if its look does not change when the entire square is rotated by a multiple of $90°$ counterclockwise around its center, nor when it is reflected across a line joining opposite corners or a line joining midpoints of opposite sides. What is the total number of possible symmetric scanning codes?

Solution. Left as an exercise. □

And finally, a problem that looks hard but is actually quite simply if you think about smaller cases and working it out.

Example 2.17

Andy the unicorn is on a number line from 1 to 2019. He starts on 1. Each step, he randomly and uniformly picks an integer greater than the integer he is currently on, and goes to it. He stops when he reaches 2019. What is the probability he is ever on 1984?

Solution. Andy only moves from $[1, 1983]$ to $[1984, 2019]$ once; there is an equal chance of landing on any of the numbers, so the probability is just $\frac{1}{36}$. □

§3 States

The bash technique that is still getting placed in the late AIME sometimes.

§3.1 Recursion

The idea is to represent the solution for a positive integer n as a_n, and then get some formula for a_n in terms of a_{n-1}, a_{n-2}, etc. and then do small cases and work up to the desired value of n. However, sometimes we need to keep track of "good" and "bad" cases, like keeping track of what the string ends with.

Example 3.1 (2008 AIME Problem 11)

Consider sequences that consist entirely of A's and B's and that have the property that every run of consecutive A's has even length, and every run of consecutive B's has odd length. Examples of such sequences are AA, B, and $AABAA$, while $BBAB$ is not such a sequence. How many such sequences have length 14?

Solution. Let a_n and b_n be the sequences of length n ending in A and B respectively. Find the recurrence relation and then solve (left as an exercise). □

§3.2 Expected Value (or Probability)

The idea here is a Markov chain, where each action moves us from one state to the next and there are a finite amount of states (so recursion is just a markov chain that always goes to smaller cases).

Some terminology:

Theorem 3.2 (States)

The success case E_S or P_S is either 0 or 1, respectively. The failure case for probability is always 0 (so it is in effect ignore); for expected value, it is also ignored. For expected value,

$$E_k = p_0 \cdot E_0 + p_1 \cdot E_1 + \cdots + p_n \cdot E_n + 1$$

where p_i is the probability to get to the state E_i, and the 1 at the end is because that took 1 more step. Also, note that none of the states can be a failure case. For probability, there is no 1 but besides that it's the same idea.

Example 3.3 (2021 AIME I Problem 12)

Let $A_1A_2A_3...A_{12}$ be a dodecagon (12-gon). Three frogs initially sit at A_4, A_8, and A_{12}. At the end of each minute, simultaneously, each of the three frogs jumps to one of the two vertices adjacent to its current position, chosen randomly and independently with both choices being equally likely. All three frogs stop jumping as soon as two frogs arrive at the same vertex at the same time. The expected number of minutes until the frogs stop jumping is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Solution. The states are:

They're all 4 away from each other. Let the expected value from this position be a.
The distances between them are $2, 4, 6$. Let the expected value from this position be b.
The distances between them are $2, 2, 8$. Let the expected value from this position be c.
One of the distances between them is 0.

$$\begin{aligned} a &= \frac{1}{4}a + \frac{3}{4}b + 1 \\ b &= \frac{1}{2}b + \frac{1}{8}a + \frac{1}{8}c + 1 \\ c &= \frac{1}{4}c + \frac{1}{4}b + 1 \end{aligned}$$

Solving we get $c = \frac{8}{3}, b = 4, a = \frac{16}{3}$, and the answer we want is $16 + 3 = 19$. □

§4 Expected Value

Theorem 4.1 (Linearity of Expectation)

Let X_1, X_2, ..., X_n be random variables (an action that has an output; like the number that shows up when you roll a fair die). Note most times that all the X_i are the same random variable, but they might not be independent. However, linearity of expectation takes care of this easily. If $E(X)$ is the expected value of the random variable X, then

$$E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$$

Note, however, this does NOT imply for example that $E(X^2) = E(X)^2$.

Whenever you think a problem (or part of the problem) might involve Linearity of Expectation, remember to keep in mind: what exactly is the random variable? and how can I get the $X_1 + X_2 + \cdots + X_n$ to represent the problem?

As an example, here's a classic:

Example 4.2

In a class of thirty students, a teacher has thirty index cards with each student's name on exactly one of them. The teacher then randomly hands every student an index card. What is the expected number of students who get an index card with their name on it?

Solution. This one's quite easy. The only action the teacher is repeating is giving a student a random index card. Of course, doing this multiple times is dependent, but we don't care about that to apply LoE. Then, the probability of one student randomly getting their index card back (when there's still 30 index cards) is $\frac{1}{30}$, so the answer is $30 \cdot \frac{1}{30} = 1$. □

Truthfully this does not come up too much, but it is good to know. In fact, most of the time expected value will just involve states.

Theorem 4.3 (Expectation as sums of probabilities)

Let X be a random variable that takes on positive integer values. Then

$$\mathbb{E}[X] = \sum_{i=1}^{\infty} P(X \geq 1)$$

Proof. This comes directly from the formula $\mathbb{E}[X] = \sum_{i=1}^{\infty} i \cdot P(X = i)$, with a double counting argument. □

This is often useful when it's easier to calculate the probability of *at least* some value being hit. As a example, consider:

Example 4.4 (HMMT 2023 C9)

There are 100 people standing in a line from left to right. Half of them are randomly chosen to face right (with all $\binom{100}{50}$ possible choices being equally likely), and the others face left. Then, while there is a pair of people who are facing each other and have no one between them, the leftmost such pair leaves the line. Compute the expected number of people remaining once this process terminates.

We present the first main claim upfirst, which itself is a little tricky: the answer for a certain specific sequence is $2X$, where X is the minimum prefix when we assign -1 to left-facing people and 1 to right-facing people.

Then to calculate $\mathbb{E}[-X]$, we may notice that the cases with $P(-X = i)$ often count $P(-X = i + 1)$ as well, etc. and so it might actually be easier to consider $\sum P(-X \geq i)$. It turns out

$$P(-X \geq i) = \frac{\binom{100}{50+i}}{\binom{100}{50}}$$

and so the summation collapses. (As an exercise, try to derive the above equation via a bijection.)

§5 Problems

Try to go through most of the AIME combinatorics problems that you can find. HMMT, CMIMC and PUMaC are also good. Make sure you can do the exercises in the handout first.